Battery Sizing for a Diesel-Hybrid System

Table of Contents

Battery Sizing for a Diesel-Hybrid System: Mechanistic vs. Neural Network Comparison

1. Introduction

This report compares a mechanistic (physics-based) battery-sizing method against a neural-network method for a hybrid PV + wind + battery + diesel power system. The system studied throughout is a **diesel-hybrid** microgrid: renewable generation and battery storage serve as much load as possible, and a diesel generator — sized by power so it can always cover peak demand — makes up the rest. Battery capacity is chosen to minimize total system levelized cost of electricity (LCOE), not to satisfy a hard reliability constraint; with diesel present, reliability is effectively guaranteed by construction, so the interesting question shifts from “does the battery keep the lights on” to “how much battery is actually worth paying for, and how well can a neural network learn that economic trade-off.”

Everything in this report is generated from real, executed code against this project’s actual data; no numbers are invented or extrapolated. A short qualitative note on a diesel-free (off-grid) version of this system appears in the discussion (§16) as motivation for keeping diesel in the design — it is not presented as a second study with its own results.

2. Research Objective and Questions

The objective is to compare a mechanistic battery-sizing method against a neural-network method for this diesel-hybrid system, and to answer:

1. How accurately can a neural network reproduce the battery capacities a mechanistic cost-minimization would have found, and how stable is that accuracy across independent train/test splits (§9, cross-validation)?
2. How does accuracy scale with training-set size (§10, learning curve)?
3. Which scenario characteristics does the model rely on most, and how much better is it than a simple baseline using only the strongest of those characteristics (§11–§12, permutation importance and a linear-regression baseline)?

4. Diesel makes a hard reliability failure essentially impossible by construction, so the question is no longer “does the AI’s battery meet the reliability target” — it is **how much does trusting the AI’s predicted battery cost economically, compared to the true cost-minimizing choice, and how much more does it lean on the diesel generator than the optimum would** (§13)?
5. How sensitive is the optimal battery size itself to the battery cost assumption (§14)?
6. How much faster is neural-network inference than a complete mechanistic battery-size search, precisely stated (§15)?

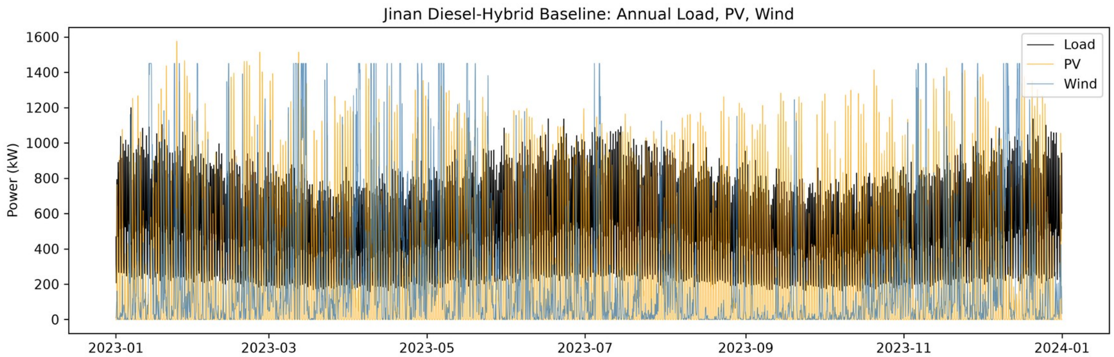
Research hypothesis, tested rather than assumed: the neural network can approximate mechanistic battery-sizing outputs closely, at a fraction of the per-scenario compute cost once trained, but its accuracy depends on the representativeness of its training data, and mechanistic verification remains useful before trusting any single prediction — even though, unlike a hard-reliability system, a diesel-hybrid system’s downside from trusting a slightly-wrong prediction is economic (a somewhat higher system cost, somewhat more diesel use) rather than a service failure. The mechanistic model is the reference method throughout: the neural network is trained on mechanistic outputs, so its performance is reported as agreement with that reference, not as validation against real, metered operational data (which this project does not have access to).

3. System Description

PV + wind generation, battery storage, a diesel generator, and an electrical load, simulated hourly over one complete local calendar year (8,760 hours, non-leap). Dispatch priority: renewable generation serves load directly first; surplus renewable energy charges the battery (any further surplus is curtailed); when renewables alone cannot cover load, the battery discharges to cover the shortfall; any load still unmet after the battery is covered by the diesel generator, up to its rated power.

The diesel generator is sized by power, not chosen by the optimization: its rated output is fixed at 1.25× the load’s achieved peak demand for every scenario, which is enough to cover the full system peak on its own if necessary. Because of this, loss-of-power-supply probability (LPSP) is effectively zero for every candidate battery size — the battery’s role is not to guarantee reliability but to reduce how much energy the diesel generator has to supply, and therefore how much is spent on both battery capital cost and diesel fuel together. The mechanistic search evaluates battery sizes from 0 to 80 modules (250 kWh each) and selects the one that minimizes total system LCOE, not the smallest one that meets a reliability target.

Site: Jinan, China (36.65°N, 117.12°E), industrial scale. PV 1,800 kWp, wind 1,450 kW (latitude-tilt fixed array + a single utility-class turbine), baseline annual load 4.5 GWh/year at a 1,200 kW peak (achieved peak ≈1,199 kW) — sized so renewable generation and load are close enough in magnitude for the diesel generator to meaningfully engage rather than sit idle (§8 shows the sensitivity of this choice).



Annual load, PV, and wind profile at the baseline case (Jinan, 2023)

4. Weather and Load Data

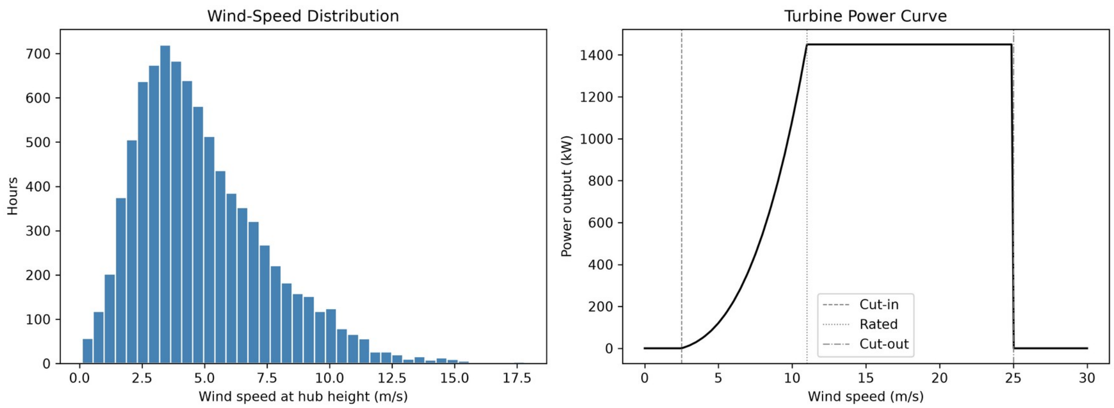
NASA POWER's hourly reanalysis product for Jinan, 2023 local calendar year, GHI/DNI/DHI/temperature/wind-speed channels, validated for unit consistency and completeness (8,760 records, no gaps). The load profile is generated by a synthetic residential-shaped load-profile model (`generate_load_profile`) scaled to the target annual energy and peak demand — this is not real metered data, and is documented as a limitation in §16.

5. Mechanistic Models: PV, Wind, Battery, Diesel

Every physics function below has hand-computed-reference unit tests.

PV: clearness index and extraterrestrial irradiance (`k_t = GHI / EGHI`), Erbs decomposition, and Liu & Jordan isotropic-sky tilted-plane transposition (`GTI = R_b·(GHI−DHI) + DHI·(1+cos β)/2 + ρ_g·GHI·(1−cos β)/2`). The main pipeline transposes using NASA POWER's measured DNI/DHI directly rather than decomposing DHI from GHI, verified numerically to be componentwise identical to the Erbs-based path for the diffuse and ground-reflected terms. NOCT cell temperature, a linear temperature-derating coefficient, and an incidence-angle modifier complete the model.

Wind: power-law height extrapolation (`v(z) = v(z_r)·(z/z_r)^α`, α = 0.14) and turbine output evaluated by linear interpolation on a discretized manufacturer-style power curve (a documented generic cubic-ramp shape, not a certified manufacturer curve).



Hub-height wind-speed distribution at the baseline site and the modelled 1,450 kW turbine's power curve

The site's hub-height wind speed is concentrated well below the turbine's 11 m/s rated speed (median ≈4 m/s), so the turbine spends most operating hours on the steep, sub-rated part of the curve rather than at its 1,450 kW plateau — wind generation at this site is real but modest relative to nameplate capacity, with PV supplying the larger share of the renewable mix.

Battery: the state-of-charge equation, including hourly self-discharge:

```
Charging:    E[t] = min(E[t-1]·(1-SDR) + η_c·E_in,t − E_out,t/η_d,
E_max)
Discharging: E[t] = max(E[t-1]·(1-SDR) + η_c·E_in,t − E_out,t/η_d,
E_min)
```

`SDR` (self-discharge rate per hour) is set to $\approx 3.4658\times10^{-5}$/hour (a typical LFP value of 2.5%/month, converted to hourly).

Diesel: rated power fixed at 1.25× the scenario's achieved peak load; fuel consumption follows a standard linear fuel curve (`F(P) = F0·P_rated + F1·P_output`) with widely-cited default coefficients; fuel is priced at 0.90 EUR/L. Capital and O&M costs are annualized alongside PV, wind, and battery costs to compute total system LCOE.

6. Neural Network Model

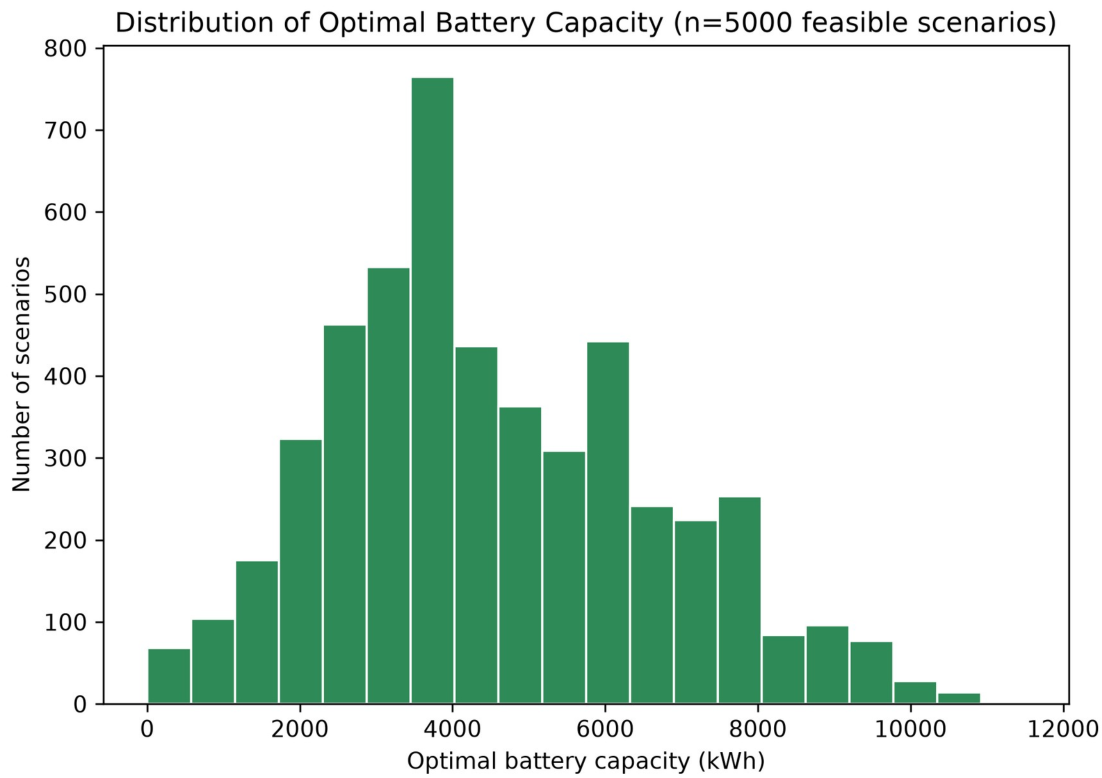
A Keras/TensorFlow MLP: three hidden layers (128/64/32 units, ReLU), 10% dropout after the first layer, ReLU output (non-negative capacity by construction), Huber loss, Adam optimizer, early stopping and best-checkpoint restoration on validation loss. Inputs are 37 engineered features per scenario (load shape, PV/wind generation statistics, net-load and deficit/surplus statistics, and the scenario's own reliability target/round-trip efficiency/usable-SOC-window); the target is the mechanistic cost-minimizing battery capacity (kWh). Every leakage-prone column (optimal capacity, module count, reference LPSP/cost/LCOE/renewable share) is excluded from the feature set and enforced by an automated guard, checked on every feature-matrix build.

A simple linear-regression baseline using only the three most important features (identified by permutation importance, §11) is also evaluated for comparison (§12), to establish how much of the neural network's accuracy comes from its top few inputs alone versus genuinely combining all 37.

7. Scenario Dataset

5,000 scenarios were sampled (PV capacity 500–3,500 kWp, wind 200–2,000 kW, combined PV+wind cap 5,000 kW, annual load 1.5–6.5 GWh/year, peak load 400–1,800 kW, reliability target 99.0–99.9%, round-trip efficiency 88–97%, usable SOC window 70–90%), each independently run through the full mechanistic pipeline at Jinan's 2023 weather. Because the diesel generator can always cover any remaining shortfall, **every one of the 5,000 scenarios (100%) is feasible** — unlike a hard-reliability system, there is no scenario a battery-plus-diesel combination cannot serve; the only question is how much battery is worth installing to minimize cost. The reliability target and round-trip efficiency remain sampled per scenario and used as model inputs, since they still affect how much the battery can usefully contribute, even though neither is a binding feasibility constraint here.

The resulting cost-minimizing battery size varies meaningfully across the dataset — mean 18.1 modules (4,522 kWh), standard deviation 8.5 modules, ranging from 0 (renewable-poor or load-heavy scenarios where diesel alone is cheapest) to 46 modules. This spread matters: because battery cost genuinely changes which size is optimal here (§14), the neural network's target is a real economic trade-off, not (as in a hard-reliability system) a value driven purely by a fixed constraint regardless of cost.



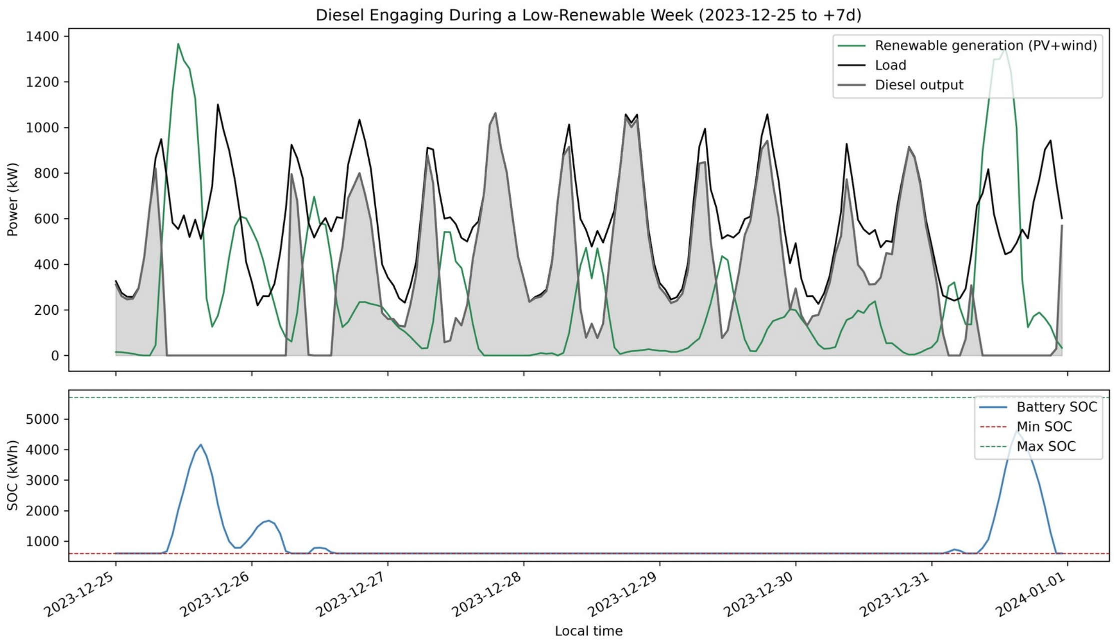
Distribution of mechanistically optimal battery capacities across the 5,000-scenario dataset

8. Baseline System: Diesel Engagement and the Renewable-Share/LCOE Trade-off

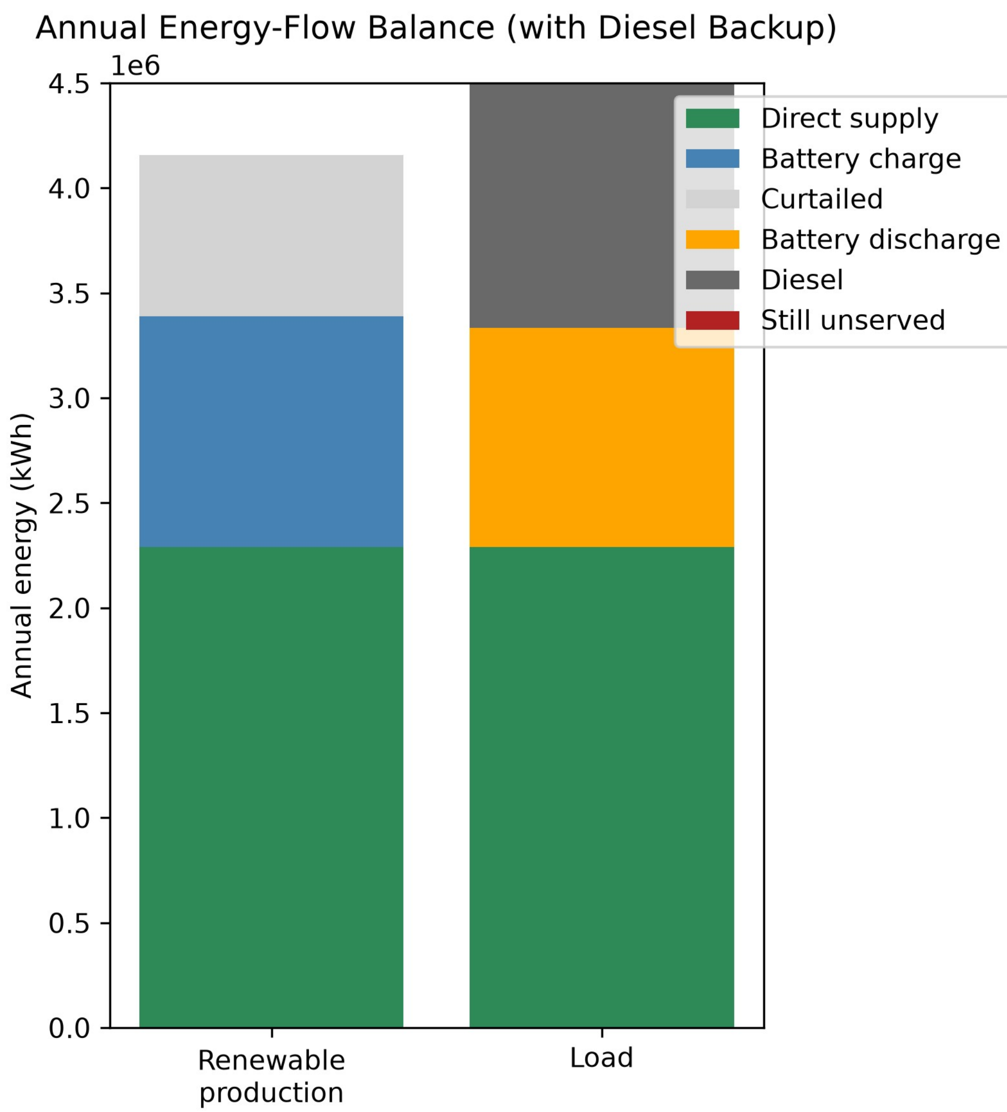
At the baseline case (§3), the mechanistic search's cost-minimizing choice is:

Metric	Value
Optimal battery	24 modules · 6,000 kWh
System LCOE at optimum	0.288 EUR/kWh
Renewable share / diesel share	74.1% / 25.9%
Curtailment (of generated renewable energy)	18.4%
Diesel operating hours	3,250 / 8,760 (37.1% of the year)
Diesel fuel consumption	683,550 L/year
Diesel capacity factor	8.87%
LPSP	0.0 (diesel guarantees reliability by construction)

The baseline load and peak (§3) were chosen deliberately so the diesel generator visibly participates rather than sitting idle: at a lower load relative to the fixed PV/wind capacity, renewable generation alone would cover the great majority of demand and diesel would barely run; at a higher load, diesel would dominate. 4.5 GWh/year at 1,200 kW peak lands in a middle range where both technologies do real, comparable work.



Diesel engaging during a low-renewable week (Dec 25-31, 2023): renewables dip well below load, battery SOC sits near its floor for most of the week, diesel fills the remaining gap



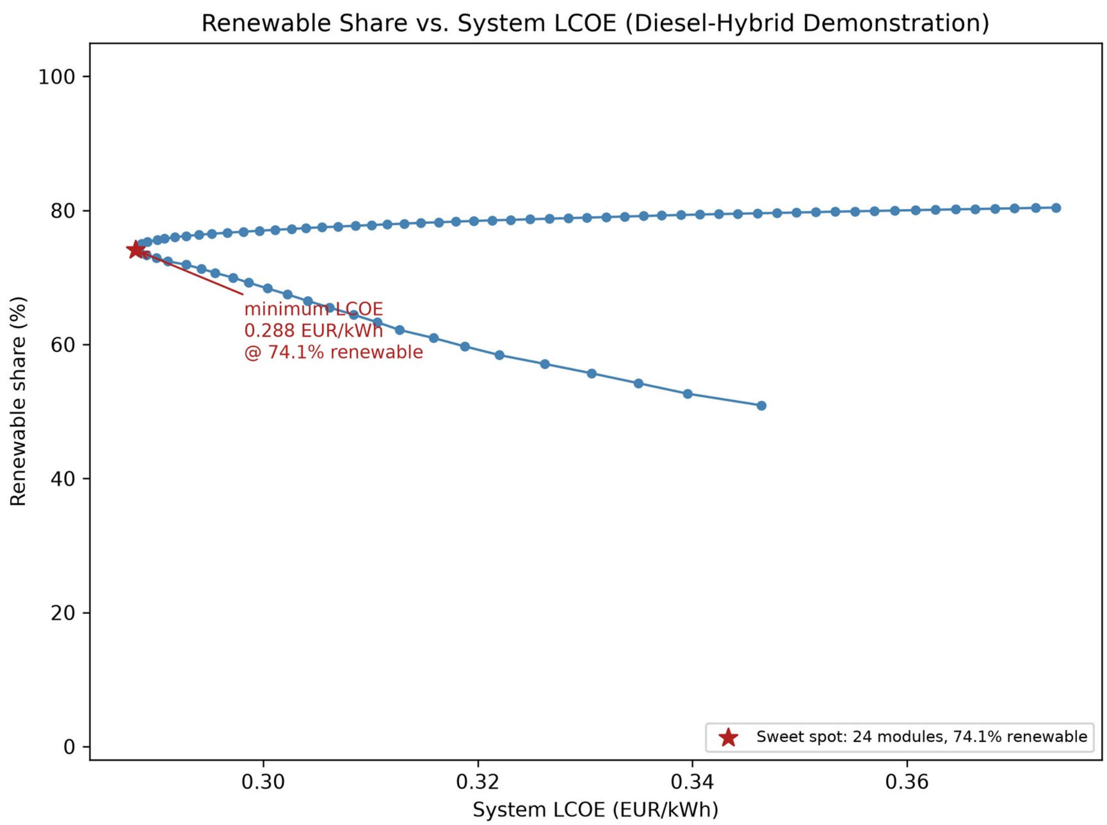
Annual energy-flow balance: renewable production (direct supply + battery charge + curtailed) and load service (direct supply + battery discharge + diesel), zero unserved

Energy balance is verified exactly every hour of the year: `load_kw` = `direct_supply_kwh` + `battery_discharge_kwh` + `diesel_output_kwh` + `still_unserved_kwh`, annual residual 0.0 kWh.

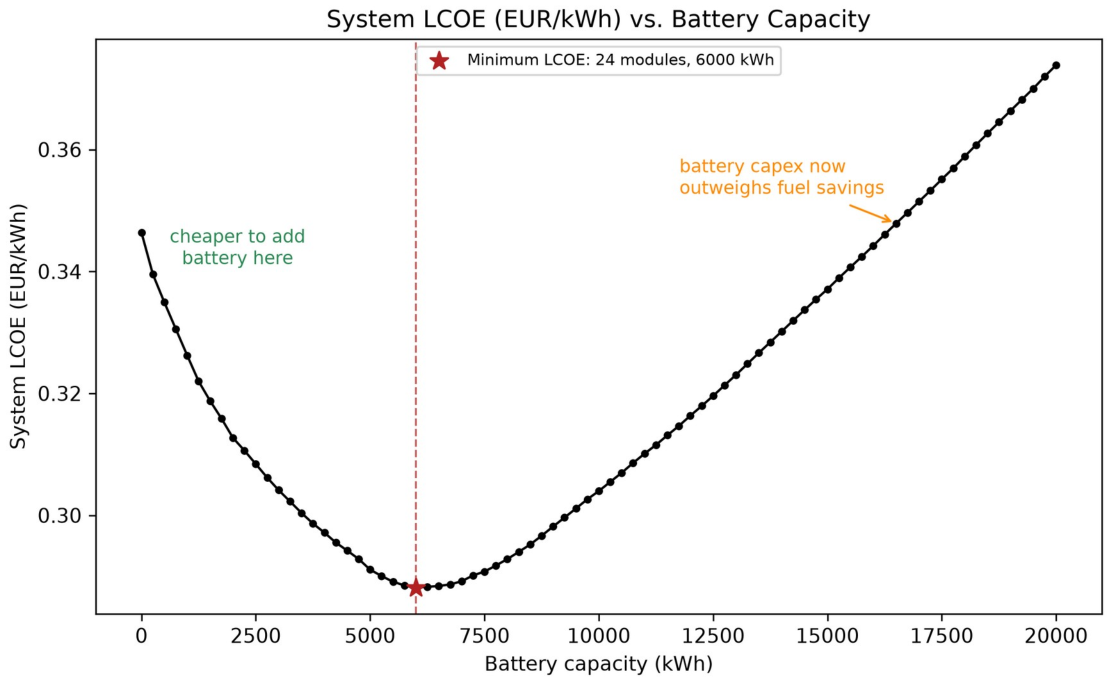
Where does more battery stop paying for itself? Sweeping the full 0–80-module candidate range at the baseline case traces a clean U-shaped (backward-bending) renewable-share-vs-LCOE curve:

Point	Modules	Battery capacity	System LCOE	Renewable share
Diesel-only	0	0 kWh	0.346	50.9%

Point	Modules	Battery capacity	System LCOE	Renewable share
(no battery)			EUR/kWh	
Cost-minimizing (the search's own optimum)	**24**	**6,000 kWh**	**0.288 EUR/kWh**	**74.1%**
Maximum tested battery	80	20,000 kWh	0.374 EUR/kWh	80.4%



Renewable share vs. system LCOE across the full 0-80 module candidate sweep, sweet spot marked at the minimum-LCOE point



Battery capacity vs. system LCOE, same sweep: a clean U-shape with the minimum at 6,000 kWh

Starting from diesel-only, each added battery module saves more in diesel fuel than it costs in capital — system LCOE falls 17% (0.346 → 0.288 EUR/kWh) as renewable share climbs from 50.9% to 74.1%. Past 24 modules, the relationship flips: each additional module's capital cost now exceeds what it saves in fuel, because the marginal battery capacity is mostly covering rarer, higher-deficit hours rather than displacing routine diesel runtime. Pushing all the way to the 80-module cap buys only another 6.3 percentage points of renewable share (74.1% → 80.4%) but costs 30% more (0.374 EUR/kWh) than the optimum.

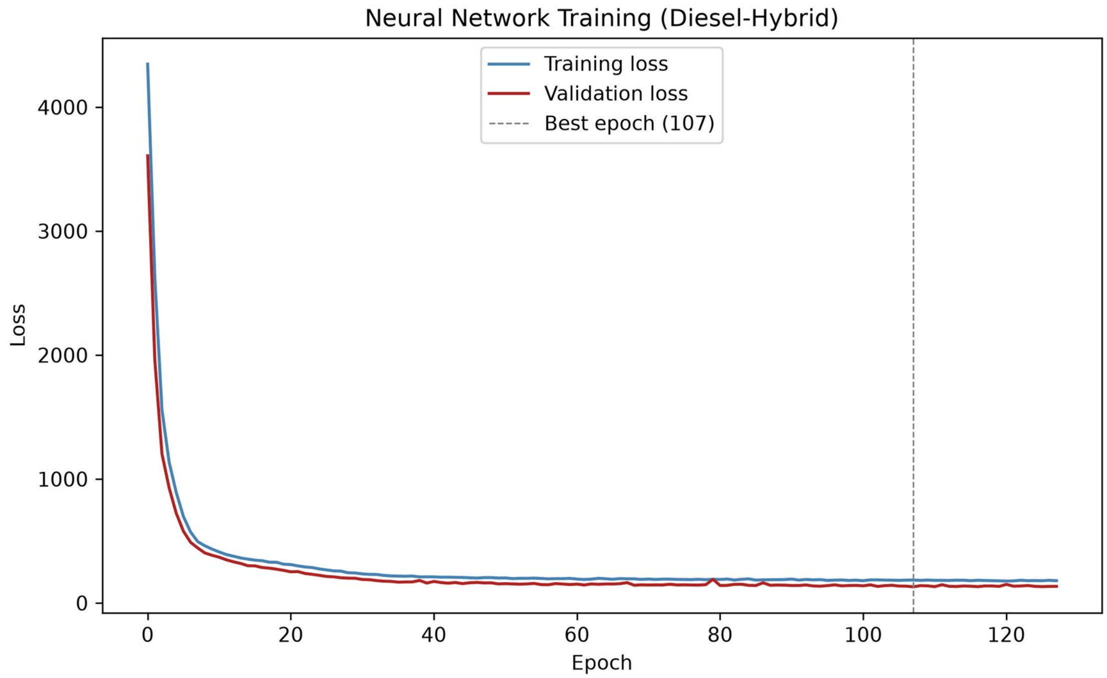
Renewable share is worth paying for up to ~74% at this baseline case; beyond that, each additional percentage point of renewable share costs markedly more per kWh delivered. This is the economic sweet spot the neural network is ultimately being asked to learn to predict, scenario by scenario, across the full sampled dataset.

9. Results: Accuracy and Cross-Validation

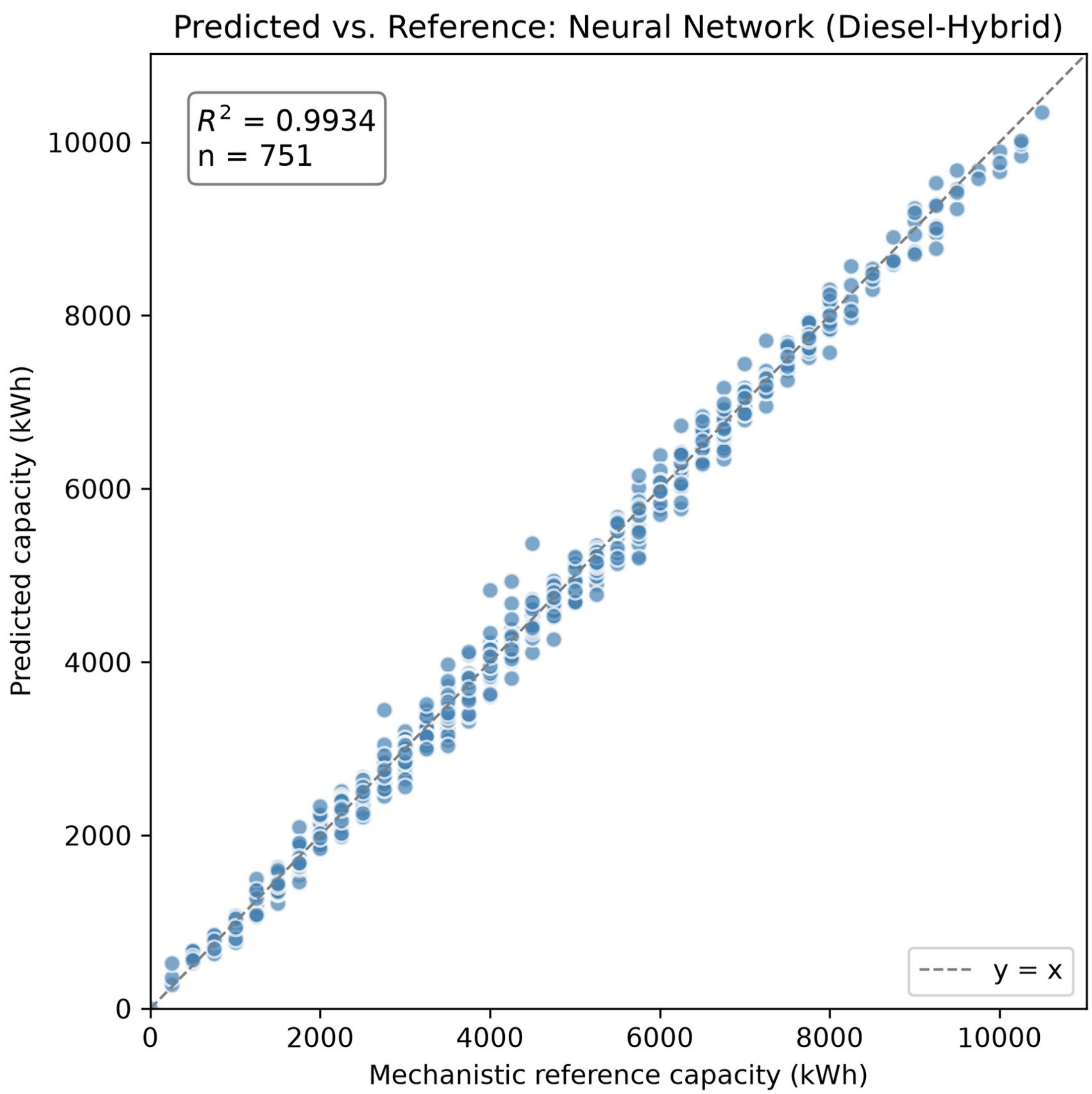
A single 70/15/15 grouped train/val/test split (seed 42, `scenario_id` as the group column — a proxy for a plain random split, since every scenario is an i.i.d. draw) gives, on the 751-scenario test split:

Metric	Value
MAE	125.9 kWh

Metric	Value
RMSE	168.2 kWh
R^2	0.9934
Within 5% of reference	79.0%
Within 10% of reference	93.9%
Within 20% of reference	98.9%



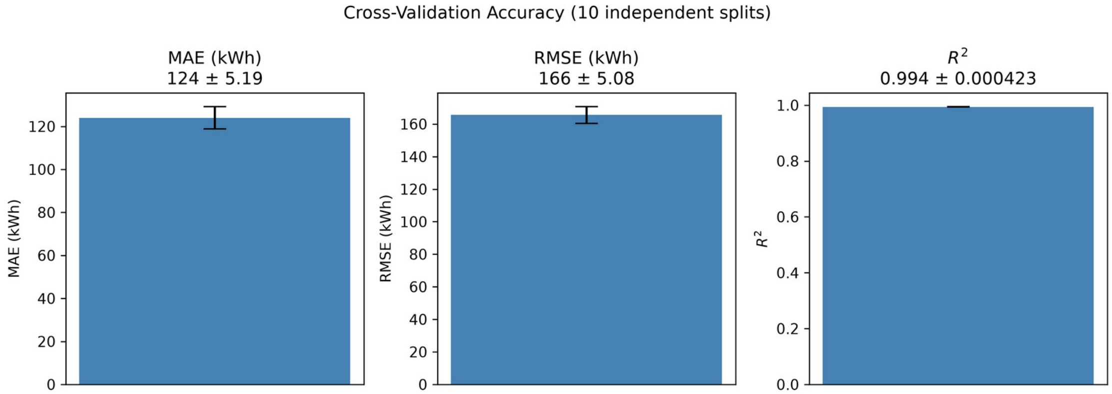
Neural network training/validation loss curves



Predicted vs. reference capacity, 1:1 line, R^2 annotated, 751-scenario test split

A single split does not, by itself, establish that this result is typical rather than a favorable draw. Ten independent repeated 70/15/15 splits (seeds 0–9, full retraining each time) give:

Metric	Mean ± Std	Min	Max
MAE (kWh)	123.9 ± 5.2	116.0	134.3
RMSE (kWh)	165.5 ± 5.1	158.6	175.4
R^2	0.9939 ± 0.0004	0.9931	0.9944



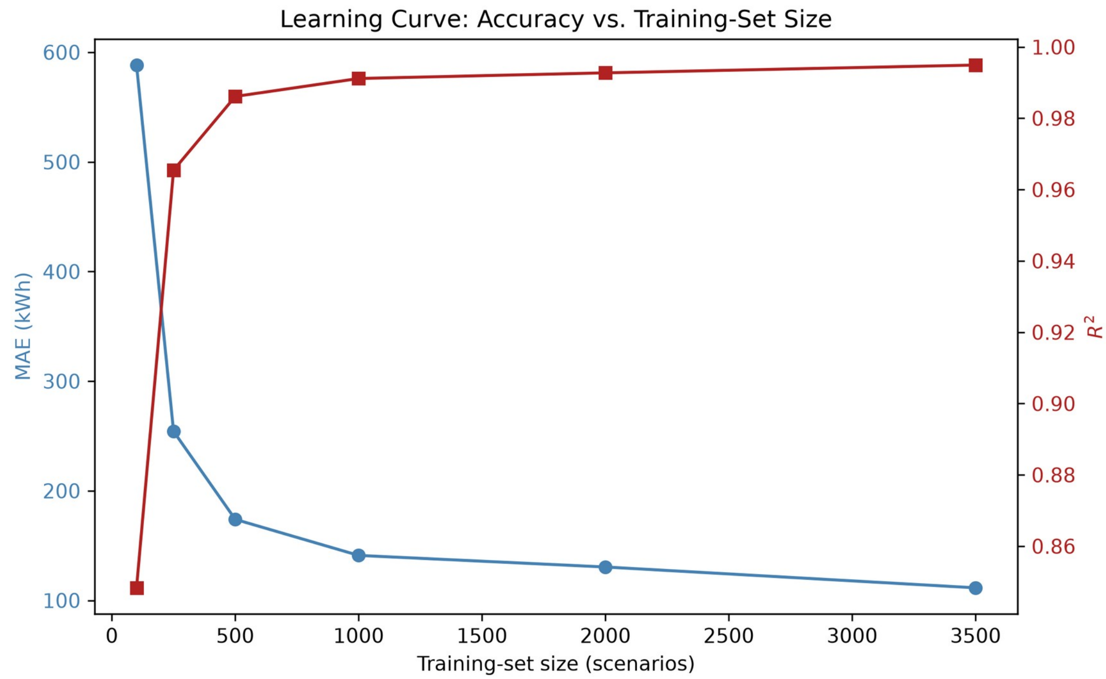
Cross-validation error bars: MAE, RMSE, and R^2 (mean ± std, 10 independent splits)

The headline seed-42 result falls squarely within this range: the reported accuracy is representative, not an outlier, and is in fact tighter (smaller standard deviation) than the equivalent cross-validated accuracy this project previously observed under a hard-reliability target — even though the diesel-hybrid target is a genuinely cost-dependent quantity rather than one driven by a fixed constraint.

10. Results: Learning Curve

Training-set size was varied from 100 to the full available 3,500-scenario training split (fixed val/test at seed 42):

Training scenarios	MAE (kWh)	R^2
100	588.3	0.848
250	254.2	0.965
500	173.9	0.986
1,000	141.1	0.991
2,000	130.5	0.993
3,500 (full)	111.5	0.995



Learning curve: MAE and R^2 vs. training-set size

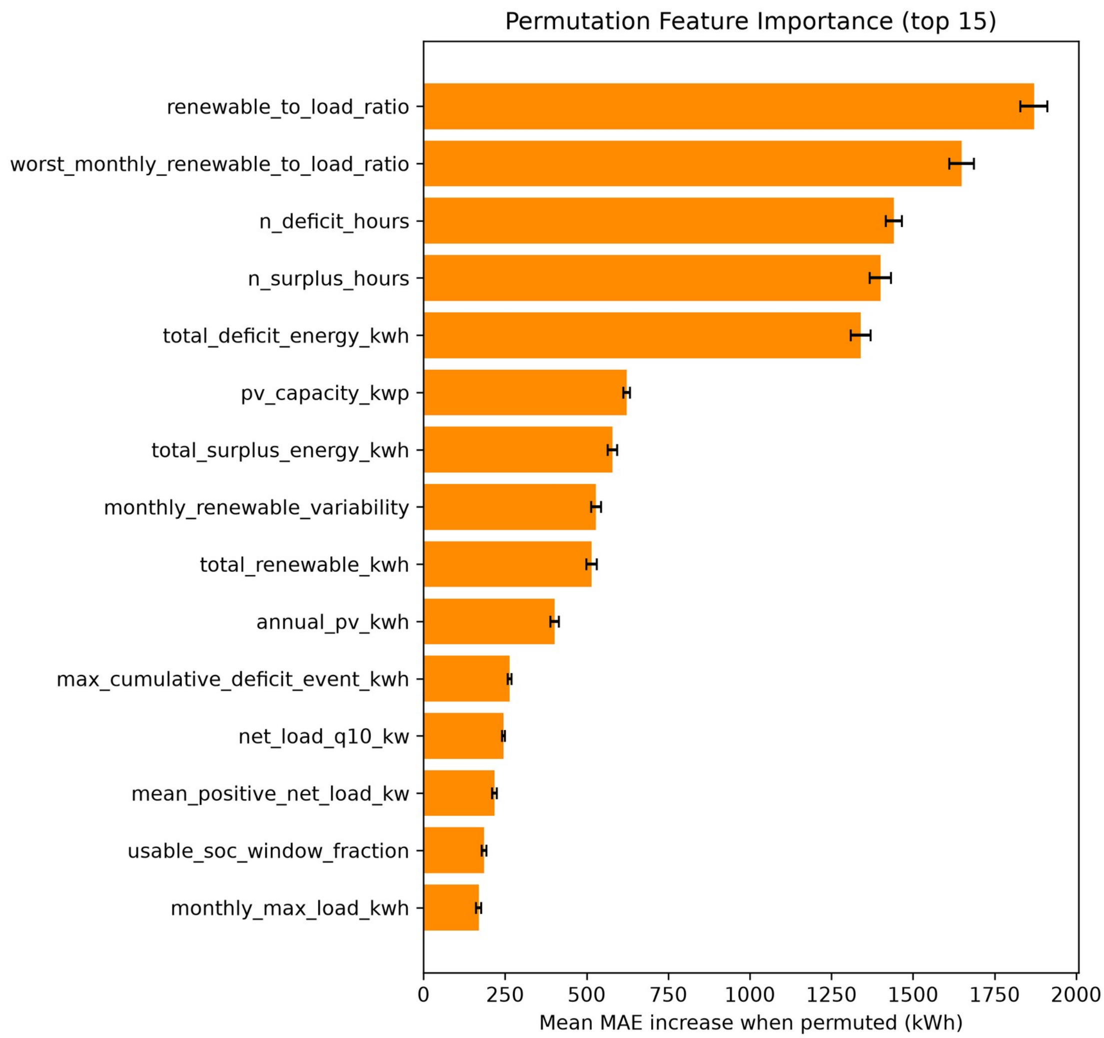
Accuracy improves sharply up to ~1,000 scenarios, then continues to improve more gradually rather than fully plateauing — the full 3,500-scenario training set still meaningfully outperforms the 2,000-scenario point (130.5 → 111.5 kWh MAE), unlike the flatter plateau this project observed for a hard-reliability target at a similar scale. This is consistent with the target itself being a more complex, genuinely cost-dependent function of the scenario inputs here: there is more signal left for additional training data to capture.

11. Results: Permutation Feature Importance

Each of the 37 input features was independently shuffled on the test split (10 repeats each) and the resulting increase in MAE recorded, using the already-trained headline model (baseline test MAE 125.9 kWh, no retraining). The five most important features are:

Feature	Mean MAE increase	% of baseline MAE
`renewable_to_load_ratio`	1,870.6 kWh	1,485%
`worst_monthly_renewable_to_load_ratio`	1,648.8 kWh	1,309%
`n_deficit_hours`	1,440.6 kWh	1,144%
`n_surplus_hours`	1,400.3 kWh	1,112%

Feature	Mean MAE increase	% of baseline MAE
`total_deficit_energy_kwh`	1,339.5 kWh	1,064%



Permutation feature importance, top 15 features by mean MAE increase

This is physically sensible: under a cost-minimization objective with diesel available, the model relies most heavily on how much renewable generation there is relative to load overall and in the worst month, and on how often and how severely renewables fall short of covering demand — exactly the quantities that determine the trade-off between battery capital cost and diesel fuel savings. This differs from what mattered most under a hard-reliability target (where the strictness of the reliability requirement itself ranked highly); here, the *economics* of covering deficits, not a pass/fail threshold, drives the ranking.

12. Results: Simple Baseline Comparison

To check how much of the neural network's accuracy comes from combining all 37 features versus relying mostly on the top few, a linear-regression model was fit using only the three most important features from §11 (`renewable_to_load_ratio`, `worst_monthly_renewable_to_load_ratio`, `n_deficit_hours`), on the same train/test split:

Model	MAE (kWh)	R^2	Within 10% of reference
Linear regression, top 3 features	1,205.1	0.453	21.8%
Neural network, all 37 features	125.9	0.993	93.9%

The full neural network's MAE is roughly ten times lower than the simple baseline's, and its R^2 is more than double. Even though the top few features dominate the permutation-importance ranking, a linear combination of just those three captures less than half the target's variance — the neural network's ability to combine all 37 features, including their non-linear interactions, accounts for the great majority of its accuracy advantage. This confirms the model is not simply reproducing what a much simpler method could already achieve from the most obviously important inputs.

13. Results: Physical Verification — Economic Cost and Reliance on Diesel

Every test-split prediction was rounded up to an installable module count (never down) and re-simulated through the full hourly mechanistic dispatch model, exactly as a deployment would have to:

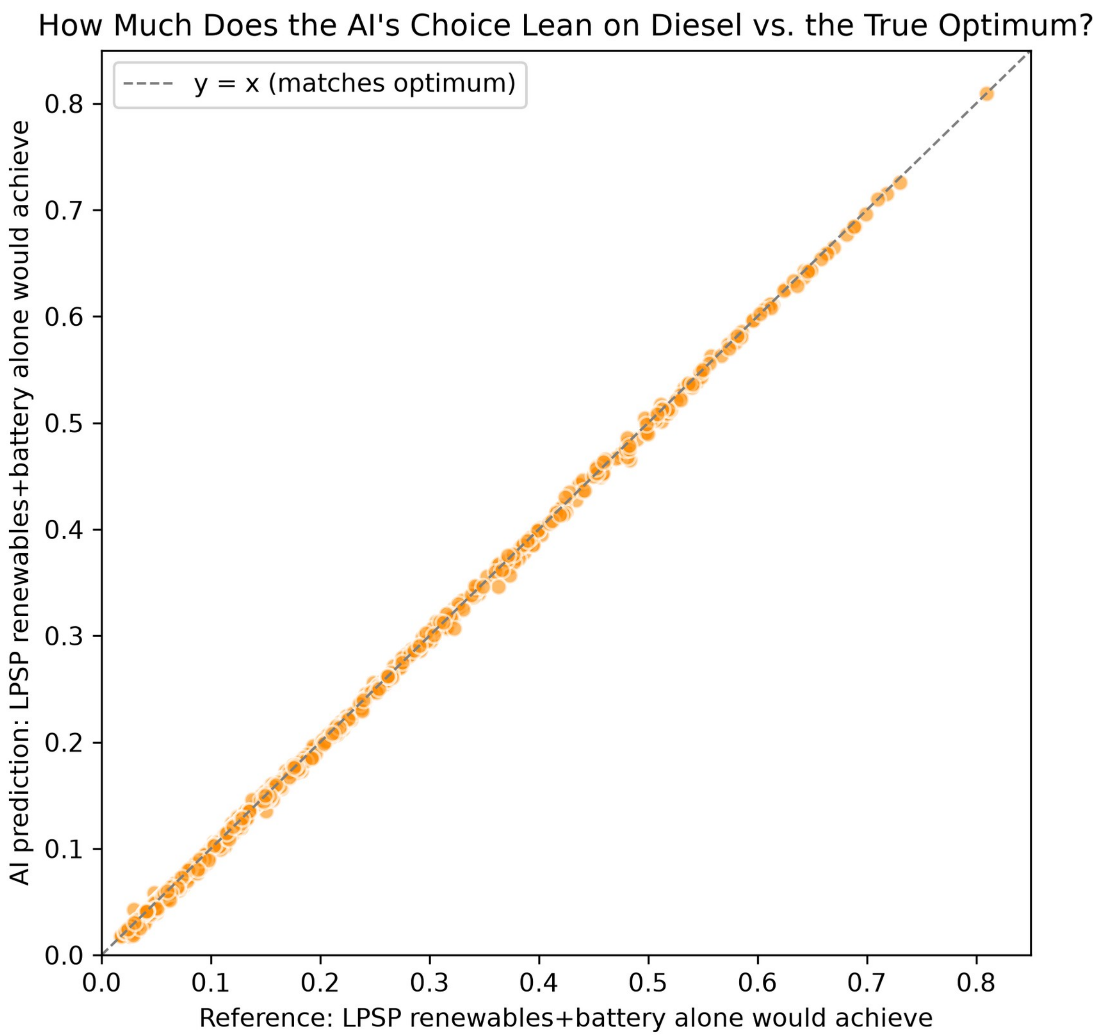
Metric	Value
Mean extra system LCOE vs. the true cost-minimizing choice	0.00013 EUR/kWh
Within 5% of the optimal system LCOE	100.0%
Undersized (predicted < reference)	6.9%
Oversized (predicted > reference)	41.4%
Mean excess capacity when oversized	96.9 kWh
Worst-case underprediction	500 kWh (two modules)

Because diesel guarantees reliability by construction, 100% of rounded predictions technically satisfy the scenario's own reliability target — this is no longer a meaningful pass/fail test (it is guaranteed for essentially any battery size, including zero), so the economic and diesel-reliance measures below are the informative ones instead.

Economic cost of trusting the AI. Rounding the AI's prediction up and installing it, instead of the true cost-minimizing battery size, raises system LCOE by an average of only 0.00013 EUR/kWh — a small fraction of a percent of the baseline system's 0.288 EUR/kWh — and every single test-split prediction lands within 5% of the true optimal system LCOE. Trusting the neural network's rounded prediction directly, without mechanistic re-verification, costs almost nothing economically in this system, in sharp contrast to a hard-reliability system where an undersized prediction can cause an outright service failure.

How much more does the AI's choice lean on diesel than the optimum would? For each scenario, the loss-of-power-supply probability that renewables and battery *alone* (without diesel) would achieve was computed for both the AI's installed capacity and the true optimal capacity — a direct measure of how much each choice would need to lean on diesel if it had to. Averaged across the test split:

Metric	Value
Mean battery-alone LPSP, AI's installed capacity	19.8%
Mean battery-alone LPSP, true optimal capacity	20.0%
Mean difference (AI minus optimum)	-0.2 percentage points
Scenarios where the AI leans *more* on diesel than the optimum would	6.9%



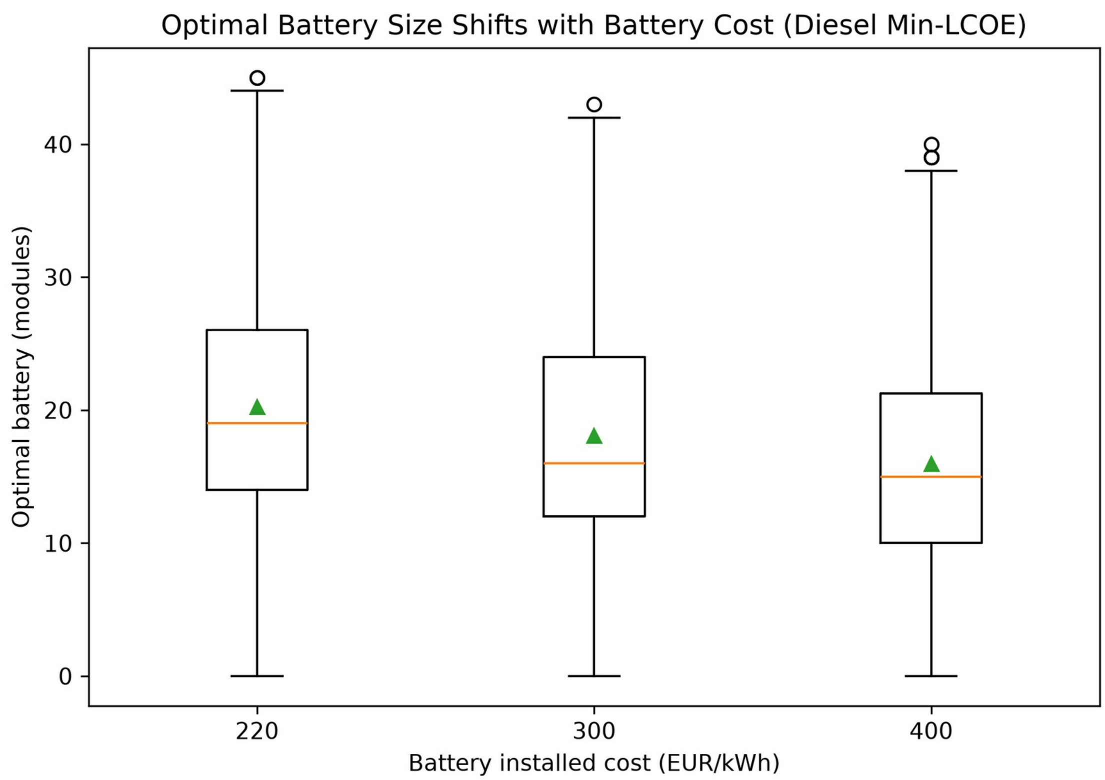
How much the AI's choice leans on diesel vs. the true optimum, per test-split scenario

On average, the AI's rounded battery choice leans on diesel very slightly *less* than the true optimum would (a difference of -0.2 percentage points — within noise of zero), and only 6.9% of test-split scenarios — closely matching the 6.9% that were undersized — see the AI leaning more heavily on diesel than the optimal choice would have. This is the direct answer to Research Question 4 (§2): trusting the AI's prediction costs almost nothing economically and shifts reliance onto diesel only in the minority of cases where the prediction undershoots the true optimum, and even then by a modest amount, not a large one.

14. Results: Cost Sensitivity

Unlike a hard-reliability system — where battery cost changes the *reported* system LCOE but never which battery size gets selected, since sizing there is driven purely by a reliability constraint — under cost-minimization with diesel available, the optimal battery genuinely depends on the battery cost assumption: a cheaper battery can profitably displace more diesel fuel, so the optimum grows; a more expensive battery displaces less, so the optimum shrinks toward diesel-only. Battery installed cost was swept across the project's documented sensitivity range (220, 300, 400 EUR/kWh, baseline 300) on a 1,000-scenario random subsample, re-running the full mechanistic search at each cost point since the selected candidate itself changes, not just its reported economics:

Battery cost (EUR/kWh)	Mean optimal battery (modules)	Std	Median
220	20.2	8.6	19
300 (baseline)	18.1	8.4	16
400	15.9	8.3	15



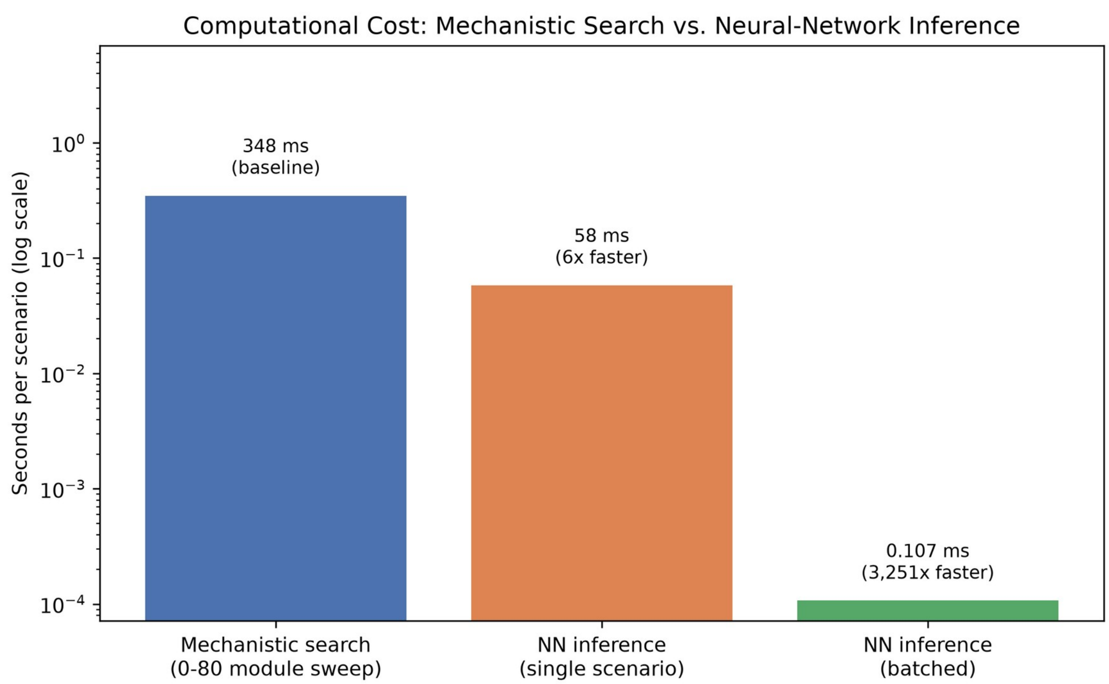
Optimal battery size shifts with battery cost — monotonic across all three price points

The shift is monotonic and consistent across the full distribution, not just at the mean: the cheaper the battery, the larger the cost-minimizing system chooses to build, because it becomes worthwhile to displace progressively more diesel fuel with capital investment. This confirms that the neural network trained at the 300 EUR/kWh baseline (§9–§12) is learning a target that is genuinely conditioned on that cost assumption — the same model would need retraining, not just re-evaluation, if the underlying battery cost assumption changed materially.

15. Computational Efficiency

Measured directly on this project's own hardware, using the diesel-hybrid dataset and the diesel-trained model:

- **Mechanistic optimization** (searching all 0–80 candidate battery sizes, each requiring a full 8,760-hour dispatch simulation followed by diesel dispatch): **348 milliseconds per scenario on average** (5,000-scenario dataset, median 338 ms, range 276–859 ms).
- **Neural-network inference, batched** (predicting all 751 test-split scenarios in one call — the realistic regime for evaluating many candidate scenarios at once): **0.107 milliseconds per scenario**, roughly **3,250× faster** than the mechanistic search, per scenario.
- **Neural-network inference, single scenario at a time** (one ad-hoc query, e.g. an interactive tool evaluating one design at a time): **~58 milliseconds on average**, dominated by per-call framework overhead rather than the arithmetic itself — still **~6× faster** than the mechanistic search, but nowhere near the batched figure.



Computational cost per scenario: mechanistic search vs. neural-network inference, single-call and batched (log scale)

This is also the direct, visual answer to the framing question raised earlier: since the neural network is trained to reproduce the mechanistic model's own output, it cannot be more *accurate* than its reference by construction (§9–§13 establish how close it gets, and how little that remaining gap costs economically) — so the comparison that actually differentiates the two methods, once accuracy is close, is computational cost, and the difference there spans more than three orders of magnitude between the mechanistic search and batched neural-network inference.

The precise, honest claim is therefore the same shape as for a hard-reliability system: the neural network's efficiency advantage is real but regime-dependent — dramatic (~3,250×) when many scenarios are evaluated together, and much more modest (~6×) for one-off single predictions, because a large fraction of the single-call latency is fixed per-call overhead rather than genuine computation. This advantage excludes the network's own one-time training cost, which the mechanistic method does not incur at all — the network is only "free" per new scenario after that upfront investment, and only pays off if many scenarios need evaluating.

16. Discussion and Limitations

- The load profile remains a synthetic, parametrically-shaped generator, not real metered data.
- The wind power curve remains a generic normalized shape, not a specific manufacturer's certified curve.
- The scenario dataset is single-location, single-weather-year (Jinan, 2023); geographic transfer and unseen-weather-year holdout remain unattempted stretch goals.
- Permutation importance (§11) measures the *trained* model's sensitivity, not a causal claim about which physical quantities matter in general.
- **A diesel-free version of this system would look very different, and is worth noting as context for why diesel is part of the design at all.** Removing diesel entirely and requiring the same 99% reliability target from renewables and battery alone, at this project's fixed PV/wind capacity, would force renewable generation to be massively over-built relative to load to survive worst-case weeks — roughly 60% of all generated renewable energy would have to be curtailed for lack of anywhere to put it, versus 18.4% in the diesel-hybrid baseline (§8). This is the practical motivation for keeping diesel in the design: it lets the system size renewables and battery for cost-effectiveness rather than worst-case survival, at the expense of some fuel consumption and emissions. This project does not develop a diesel-free version as a parallel study with its own dataset or trained model — the observation above is qualitative context, not a separate set of results.

- The cost-sensitivity result (§14) means the trained model is specific to the baseline 300 EUR/kWh battery cost assumption; a materially different cost assumption would shift the target distribution and likely require retraining, not just re-evaluation.
- Diesel fuel price sensitivity was not explored — only battery cost was swept (§14); this is a genuine limitation, noted rather than glossed over.

17. Conclusion

For this diesel-hybrid system, the neural network reproduces mechanistic cost-minimizing battery-sizing decisions closely ($R^2 = 0.994 \pm 0.0004$ across ten independent splits) and does so at a large, precisely-quantified computational advantage over the full mechanistic search — dramatic (~3,250×) when evaluating many scenarios at once, more modest (~6×) for one-off queries. Because reliability is guaranteed by the diesel generator's power sizing, the meaningful risk from trusting the AI's prediction is economic rather than a service failure: rounding its prediction up and installing it directly raises system cost by a negligible 0.00013 EUR/kWh on average and shifts reliance onto diesel by only 6.9% of scenarios (and even then modestly), a materially lower-stakes outcome than an undersized battery in a hard-reliability system. The model's accuracy comes substantially from combining all of its input features rather than just the top few — a simple linear baseline using only the three most important features achieves an R^2 of 0.453, roughly half the neural network's. Battery cost genuinely changes which battery size is optimal here (mean 20.2 modules at 220 EUR/kWh down to 15.9 at 400 EUR/kWh), unlike a hard-reliability system where cost affects economics but not the chosen size — confirming that this system's battery-sizing problem is a genuine economic optimization, and that the trained model's target is conditioned on the battery cost assumption it was trained under.